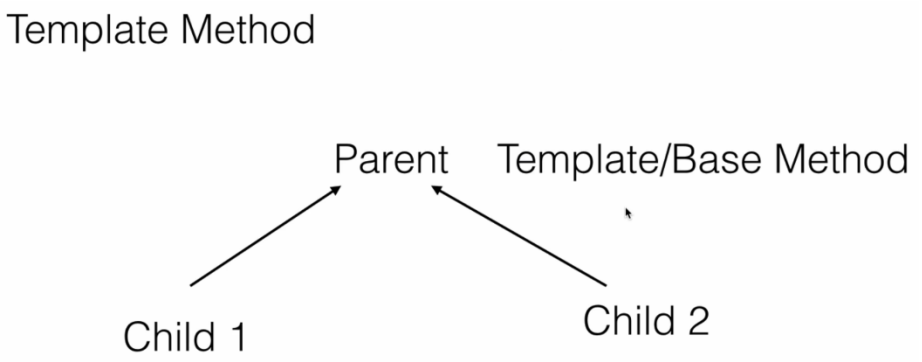
4-Template Method

- Template Method pattern is a behavioral pattern and as the name itself says it provides a base template method.
- យើងគិតវាដូចជារូបមន្ត ចឹងតើវាត្រូវការដំបូងធ្វើអីគេពីមុន។ យើងប្រើវាដើម្បីកាត់បន្ថយកូដ ដដែលៗ ហើយប្រើប្រាស់រូបមន្តដូចគ្នាៗ។



The **Template Method pattern** is a behavioral design pattern that defines the **skeleton of an algorithm** in a base class, while allowing subclasses to **override specific steps** without changing the overall structure.

🧠 Simple Idea

Think of it like a **recipe**:

- The recipe (template) defines the **steps in order**
- But some steps can be customized depending on the dish

👉 Example:

> Making tea vs coffee
> Both follow the same process:

1. Boil water
2. Brew
3. Pour into cup
4. Add extras

But:

- Tea → steep tea leaves
- Coffee → brew coffee grounds

📐 Structure

- **Abstract Class (Template)**
  - Defines the template method (final method)
  - Contains steps (some implemented, some abstract)
- **Concrete Class**
  - Implements the variable steps

🔑 Key Points

- The **template method** is usually `final` → prevents changing algorithm structure
- Subclasses only override specific steps
- Promotes **code reuse** and **consistency**

🧩 When to Use

Use Template Method when:

- You have a **fixed algorithm structure**
- Some steps **vary across implementations**
- You want to **avoid duplicate code**

⚖️ Advantages

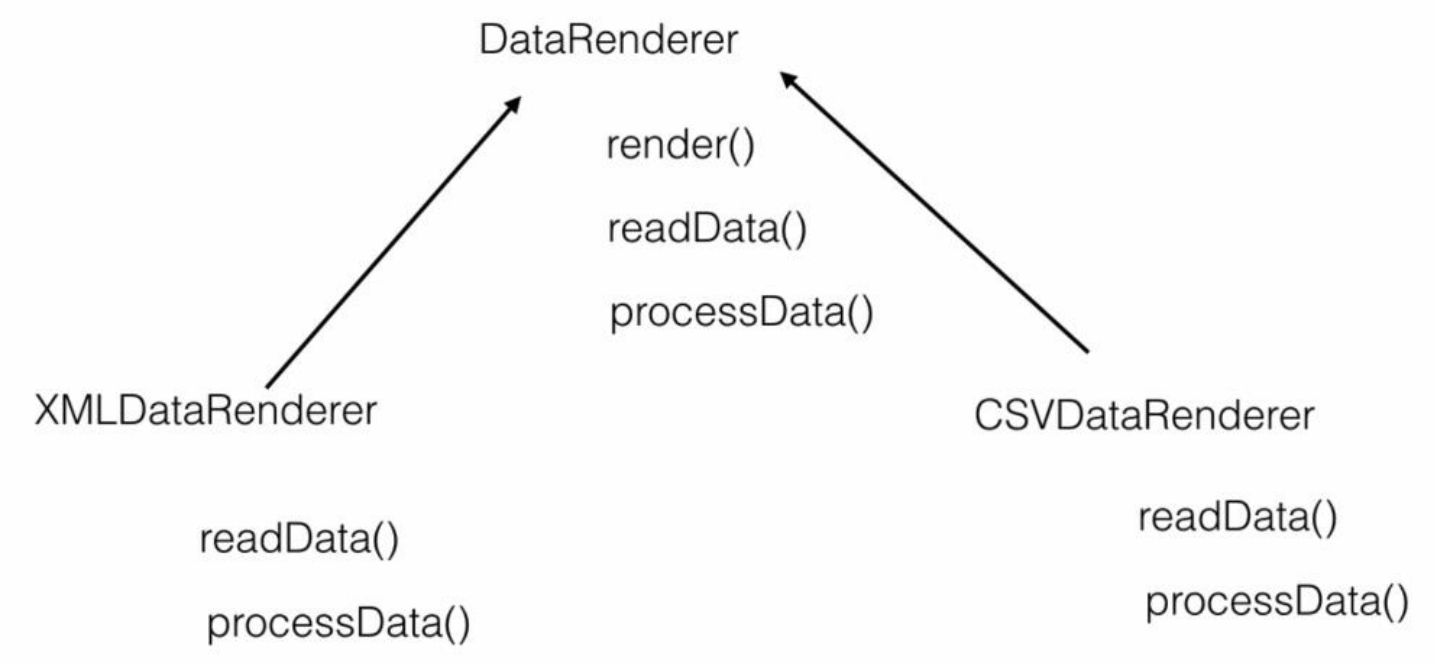
✔ Reuse common logic
✔ Enforces consistent workflow
✔ Easy to extend

⚠️ Disadvantages

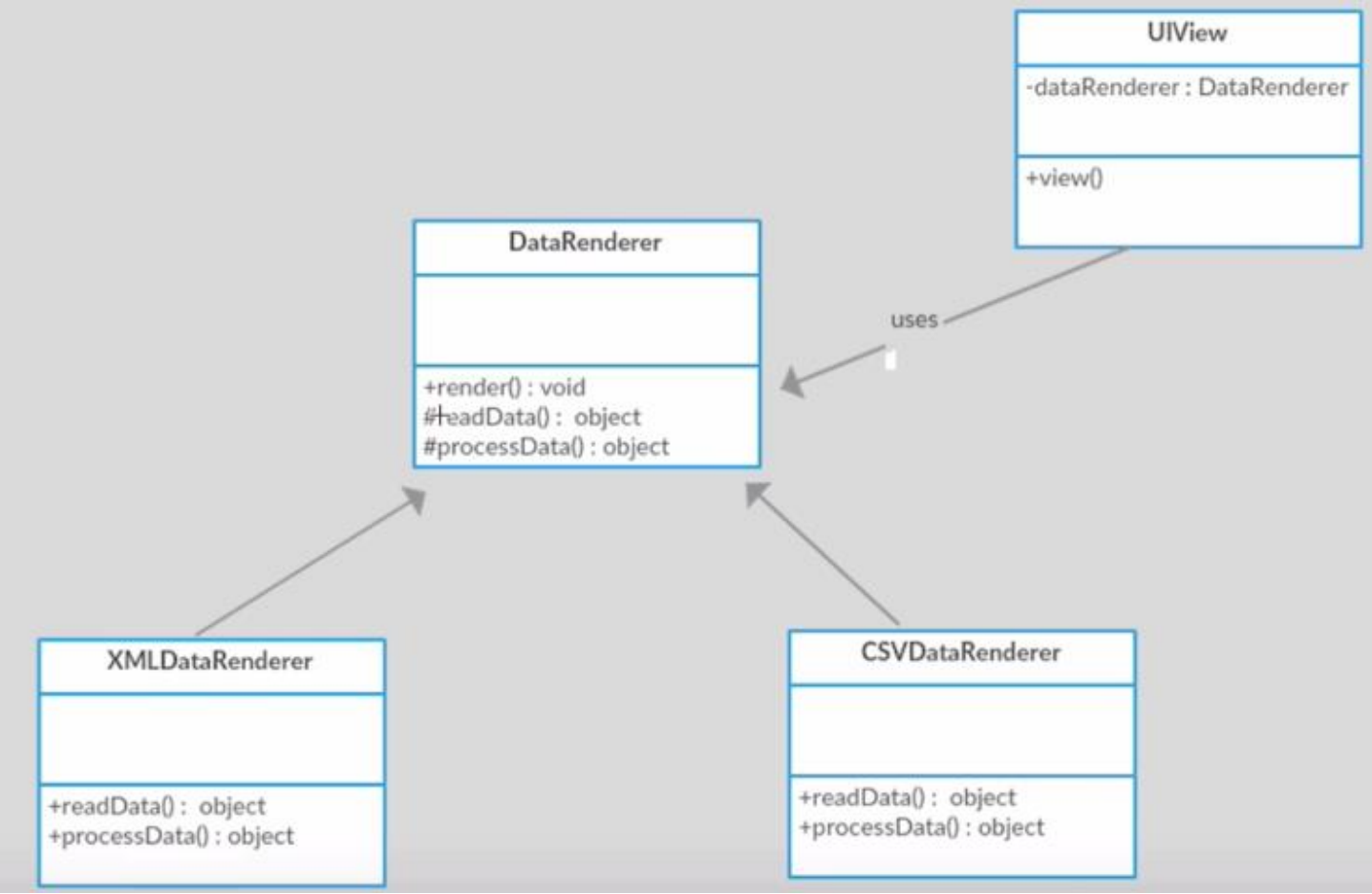
❌ Can increase number of classes
❌ Harder to change overall algorithm later

🧠 Real-World Examples

- Java:
  - `AbstractList` in Java Collections
  - `HttpServlet` (`doGet`, `doPost`)
- Frameworks:
  - Spring lifecycle methods



Example:



DataRenderer : ការបង្ហាញទិន្នន័យពី XML or CSV file

គិតថាដើម្បី Display data from xml or csv វាមានក្បួនតែមួយទេ គេធ្វើការ ReadData then ProcessData។ ចឹងវាធ្វើការដូចគ្នាគ្រាន់តែរបៀបធ្វើការវាខុសគ្នាដោយសារតែ វាជា file ផ្សេងគ្នា

Note:

- The keyword abstract in Java is used to define incomplete things—classes or methods that are meant to be extended and completed later, not used directly.
- យើងដាក់ Keyword abstract ដើម្បីបញ្ជាក់ប្រាប់ថា class នឹងមិនអាចបង្កើត object from abstract បានទេ។ តែយើងអាចបង្កើតចេញពី subclass បាន

🔹 1. Abstract Class

```java
abstract class Animal {
    abstract void makeSound();
}
```

👉 What it means:

- You **cannot create an object** of this class

```java
new Animal(); ❌
```

- It is meant to be a **base (parent) class**
- It can contain:
  - abstract methods (no body)
  - normal methods (with body)

🔹 2. Abstract Method

```java
abstract void makeSound();
```

👉 What it means:

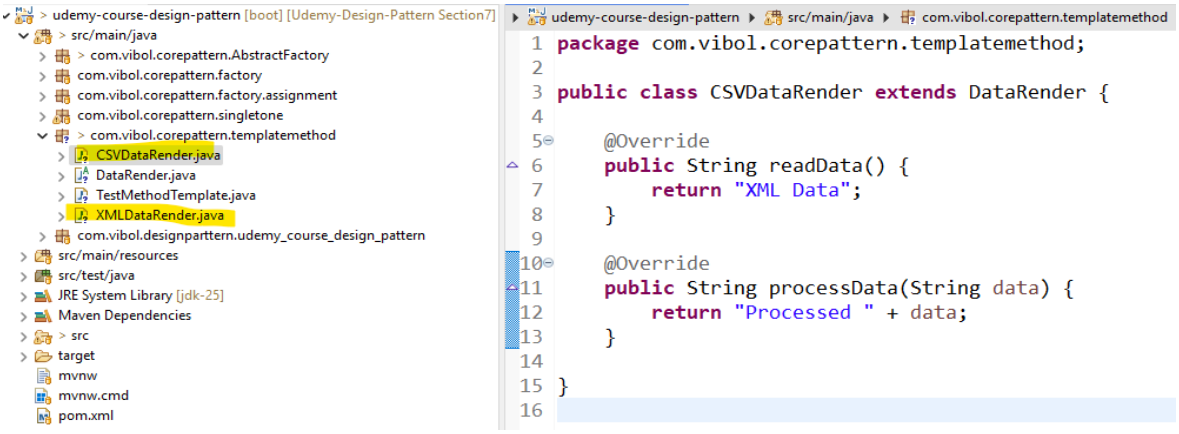
- No implementation (no `{}`)
- Child classes **must implement it**

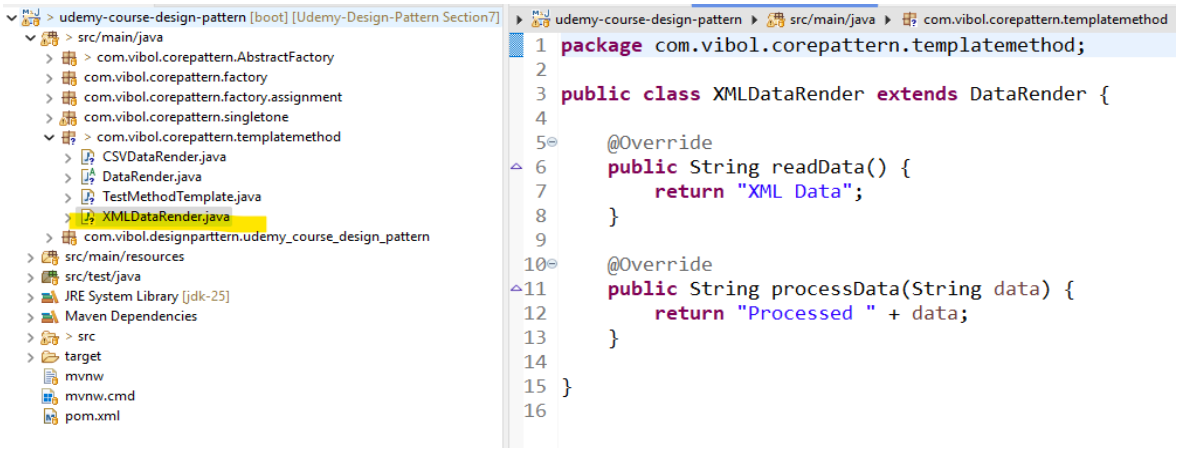
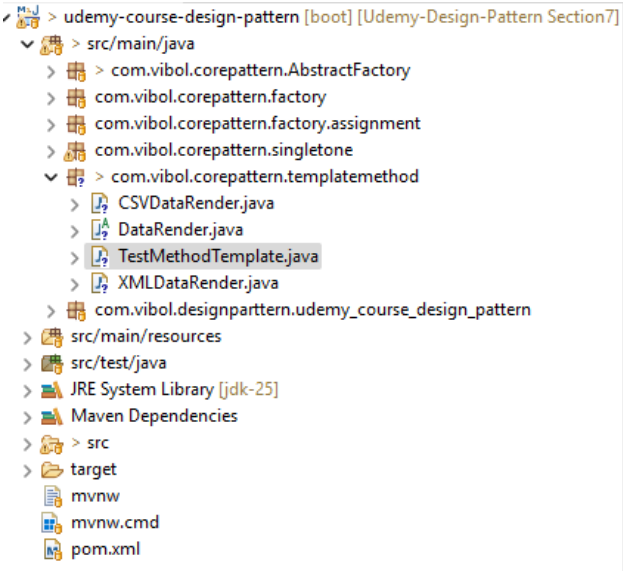
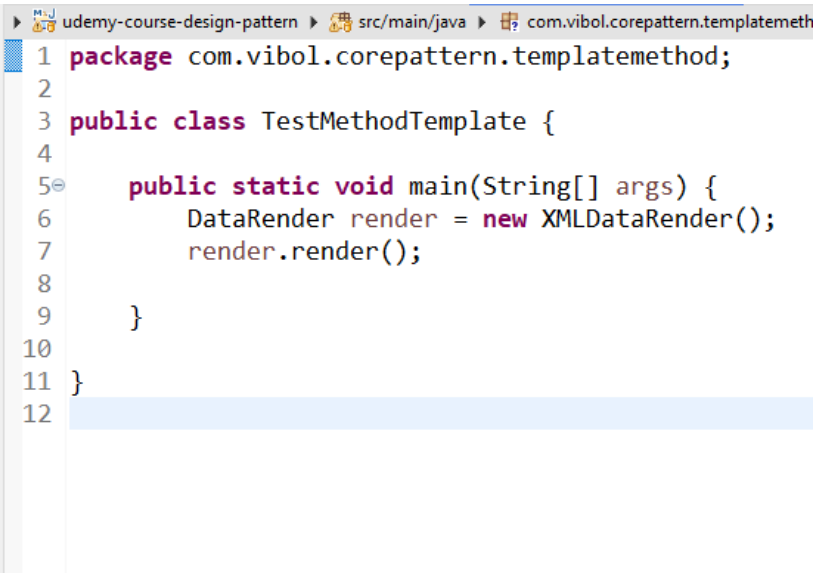
Step by Step Create Render Example:

1- Create abstract class to create process inside it

```java
package com.vibol.corepattern.templatemethod;

public abstract class DataRender {

    public void render() {

        String data = readData();
        String processData = processData(data);
        System.out.println(processData);

    }

    public abstract String readData();
    public abstract String processData(String data);

}
```

2- Implement class CSV and XML





3- Implement class